

“How Shoe Inserts Affect Lower Extremity Squat Mechanics”

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Introduction

Squatting is a common functional movement that requires coordinated control of the ankle, knee, and hip joints to maintain balance and support body weight. Squatting requires coordinated ankle, knee, and hip mechanics to maintain balance and support body weight. Orthopedic inserts are commonly used to improve foot alignment and alter force distribution through the lower extremity, potentially affecting squat biomechanics [1–5]. The purpose of this study was to determine whether orthopedic inserts alter lower-extremity joint angles and torques during squatting compared to flat shoes alone. We hypothesized that inserts would modify lower-extremity mechanics by changing underfoot support and loading patterns.

Because changes at the foot may affect ankle, knee, and hip mechanics, inserts may influence joint loading and movement strategy during weight-bearing tasks such as squatting. Center of pressure (CoP), joint angles, and joint torques can be used to evaluate postural stability, movement patterns, and joint loading during squat [6]. The aim of this study is to determine whether orthopedic inserts change lower-extremity biomechanics during a squat compared with flat shoes alone. Specifically, this study examines ankle, knee, and hip joint angles and torques across footwear conditions. We hypothesize that inserts will alter lower-extremity joint mechanics by changing underfoot support and force distribution during squatting.

Methods

Subjects and Characteristics

Four healthy adult participants participated in this study ($n = 4$; average age = 22 years; average height = 164.86 cm; average body mass = 67.38 kg; sex distribution = Male: (2) 50%, Female: (2) 50%). Each participant performed the same squat task under two footwear conditions: flat shoes only and flat shoes with inserts. Each participant completed three trials per condition, for a total of six squat trials. A within-subject design allowed each participant to serve as their own comparison across conditions.

Motion Capture Set Up

Motion capture data were collected using a six-camera motion capture system [Fig 1]. Subjects began in a neutral standing position, stepped onto the force plate with both feet, descended into a squat, held the squat for five seconds, then ascended and returned to standing. 52 markers were added in post-processing using DeepLabCut and selected frames from all six camera views. These markers were used to track lower-extremity motion and calculate hip, knee, and ankle joint angles.

Source for Ground Reaction Force (GRF) Data

A template .mot file was used to produce synchronized GRF data for each trial. Because the available .csv data contained only the center of pressure values, a single force vector was produced between the feet. This force was divided equally between the two feet and mapped to midfoot markers, so each foot received half of the recorded force at its center point. This is because in theory during a squat the forces produced and acted on are equal for each leg.

OpenSim Musculoskeletal Model

The “Generic OpenSim Model” from OpenSim was used as the initial musculoskeletal model. Each subject’s model was scaled once using one .trc trial file because body proportions were assumed to remain constant across trials and conditions. Scaling used frames near the beginning or end of the trial when the participant was standing neutrally to reduce marker motion artifacts.

Activity Cycle Definition

The activity cycle for this study is defined as the time directly before the squat descent begins (0-15%), through the isometric, 5-second hold (15-90%), and finalizes at the moment of complete ascension (90-100%) [Fig 2].

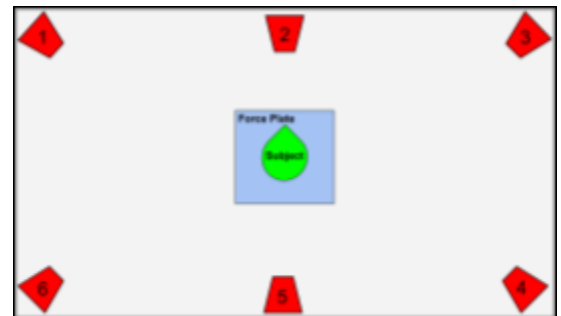


Figure 1. Motion Capture environment setup.

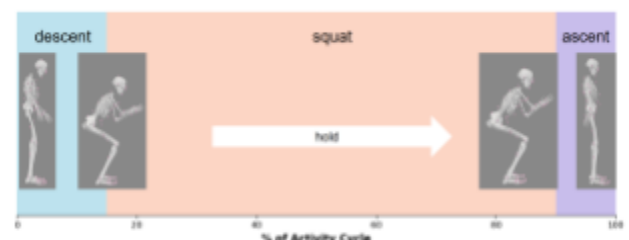
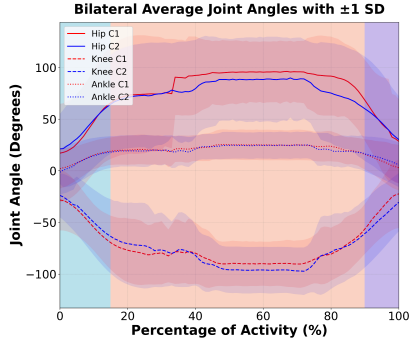


Figure 2. Activity cycle visualization.

Results



Joint	Metric (Degrees)	C1 (No Insert) Mean $\hat{A} \pm SD$	C2 (Insert) Mean $\hat{A} \pm SD$	Mean Diff	t-statistic	p-value * = $p < 0.05$
Hip	Max Angle	$92.75^{\circ} \hat{A} \pm 18.92^{\circ}$	$91.38^{\circ} \hat{A} \pm 32.45^{\circ}$	-1.37°	0.195	0.8579
Hip	Min Angle	$19.68^{\circ} \hat{A} \pm 35.45^{\circ}$	$18.40^{\circ} \hat{A} \pm 35.76^{\circ}$	-1.29°	1.521	0.2256
Knee	Max Angle	$-20.29^{\circ} \hat{A} \pm 27.35^{\circ}$	$-21.12^{\circ} \hat{A} \pm 21.04^{\circ}$	-0.82°	0.261	0.811
Knee	Min Angle	$-91.89^{\circ} \hat{A} \pm 19.43^{\circ}$	$-98.29^{\circ} \hat{A} \pm 26.06^{\circ}$	-6.40°	1.17	0.3264
Ankle	Max Angle	$23.23^{\circ} \hat{A} \pm 14.21^{\circ}$	$25.38^{\circ} \hat{A} \pm 13.83^{\circ}$	2.15°	-1.827	0.1651
Ankle	Min Angle	$-3.27^{\circ} \hat{A} \pm 19.72^{\circ}$	$-1.37^{\circ} \hat{A} \pm 18.67^{\circ}$	1.90°	-0.804	0.4801

Figure 3. Joint Angles for Hip, Knee, and Ankle Table 1. Inverse Kinematics joint angles statistics (n = 4). Table 1. Inverse Kinematics joint angle statistics (n = 4). See appendix for individually plotted hip knee and joint angles.

The inverse kinematics results indicate that the use of an insert (C2) did not produce statistically significant changes in the lower limb joint angles compared to the no-insert condition (C1). No significant differences in ankle, knee, or hip joint angles were observed between conditions ($p > 0.05$). Both conditions demonstrated similar squat mechanics characterized by increased ankle dorsiflexion, as well as knee and hip flexion during descent followed by extension during ascent.

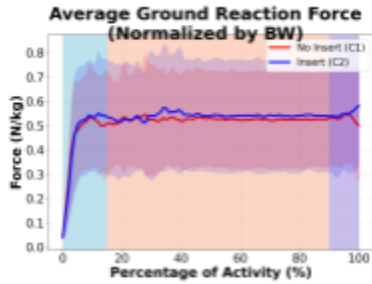
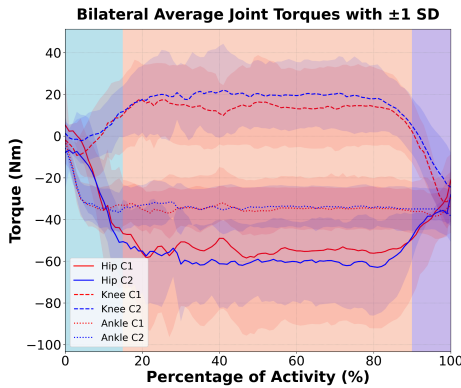


Figure 4. Average GRF for both conditions over an activity cycle.

The ground reaction forces behave similarly throughout the activity, beginning with increasing force as the trials begin. The hold and ascending phases both remain linear, sitting at $y = 0.5$ N/kg. However, it is important to note that the GRF forces have a large standard deviation, likely due to postural sway as the subject holds the squat.



Joint	Metric (Nm)	C1 (No Insert) Mean $\hat{A} \pm SD$	C2 (Insert) Mean $\hat{A} \pm SD$	Mean Diff	t-statistic	p-value * = $p < 0.05$
Hip	Max	$-47.03 \hat{A} \pm 33.43$	$-51.49 \hat{A} \pm 16.96$	-4.46	0.528	0.6339
Hip	Min	$-47.03 \hat{A} \pm 33.43$	$-51.49 \hat{A} \pm 16.96$	-4.46	0.528	0.6339
Knee	Max	$8.24 \hat{A} \pm 18.61$	$12.94 \hat{A} \pm 15.40$	4.7	-2.32	0.1031
Knee	Min	$8.24 \hat{A} \pm 18.61$	$12.94 \hat{A} \pm 15.40$	4.7	-2.32	0.1031
Ankle	Max	$-33.94 \hat{A} \pm 11.38$	$-33.31 \hat{A} \pm 10.94$	0.63	-1.464	0.2393
Ankle	Min	$-33.94 \hat{A} \pm 11.38$	$-33.31 \hat{A} \pm 10.94$	0.63	-1.464	0.2393

Figure 5. Joint Torques for Hip, Knee, and Ankle Table 2. Inverse Dynamics joint torque statistics (n = 4). See appendix for individually plotted hip knee and joint torques

The inverse dynamics results indicate that the use of an insert (C2) did not produce statistically significant changes in the lower limb joint torques compared to the no-insert condition (C1). The associated p-values for all metrics remain well above the 0.05 threshold. The t-test results and high standard deviation suggests this variation is likely due to individual subject variability rather than a systematic effect of the inserts.

Equations of Motion Analysis

Primarily, assumptions need to be made about the squatting motion and forces applied. The peak vertical GRF occurs during the isometric hold at the bottom of the squat, meaning there is no linear acceleration, angular acceleration, ground reaction moment, or major x-direction ground reaction force during the static hold. Average ϕ_{foot} was calculated to be 170.92° from OpenSim marker locations on the toes and heel. Then, using relative equations, ϕ_{tibia} and ϕ_{thigh} were

calculated to have average values of 57.69° and 138.92° , respectively. Segment masses can be determined from OpenSim model data. Then, simplifying the trigonometry and plugging in assumptions the following equations for torque emerge: $M_A = 0.9875 R_{GZ} (L_F + d_{GRF} - d_{COM,F}) - 9.687 m_F (L_F - d_{COM,F})$, $M_K = -M_A + 0.5345 R_{GZ} (L_T - 2d_{COM,t}) + 5.243 m_t d_{COM,t} - m_F (L_t - 2d_{COM,t})$, $M_H = -M_K - 0.7538 R_{GZ} L_T + 7.394 [(m_F + m_t) L_T + m_T d_{COM,T}]$.

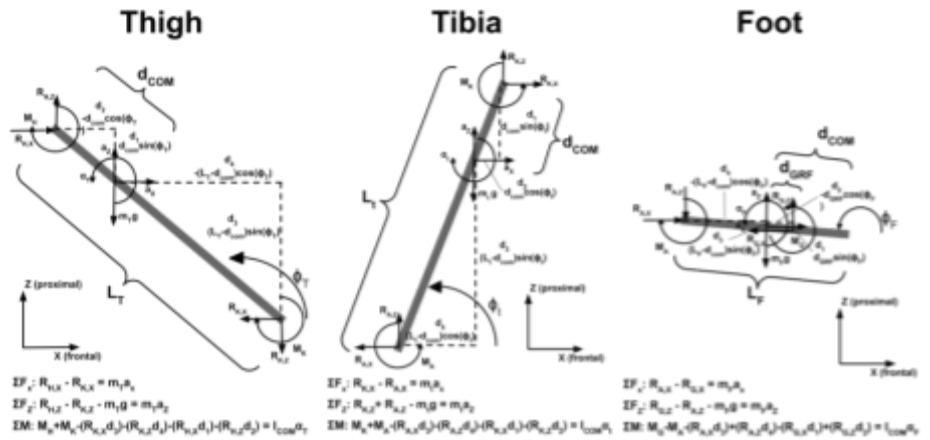


Figure 6. Free Body Diagram (FBD) of the foot, tibia, and thigh during the moment of highest GRF (isometric hold).

Subj. #	M_A (Nm)	M_K (Nm)	M_H (Nm)
1	-2.747	6.253	26.125
2	-2.787	6.531	27.293
3	-1.603	3.703	15.498
4	-1.373	3.185	13.331

Table 3. Ankle, Knee, Hip Torque

Using the equations of motion from the FBD creates torques that differ from those calculated earlier. This is likely due to the assumed positive directions in both models. Both models have similar magnitudes for each respective torque, with the ankle applying the least, followed by the knee, and the most torque applied by the hip. This is biomechanically logical as larger, more powerful muscles act on those joints in the same order.

Knee Muscular Analysis

The main extensor of the knee joint is the quadriceps, which works in opposition to the main flexor of the knee, the hamstring. Joint torque in the knee did not change significantly between conditions [Table 3]. The squatting motion is almost entirely dominated by the quadriceps in order to control the eccentric descent, maintain a steady isometric hold at the bottom of the squat, and then contract to lift the subject during the concentric ascent. The hamstring would likely be largely inactive during the descent and ascent portions of the squat, however it is likely to be active during the isometric hold as a stabilizing force against the quadriceps.

Discussion

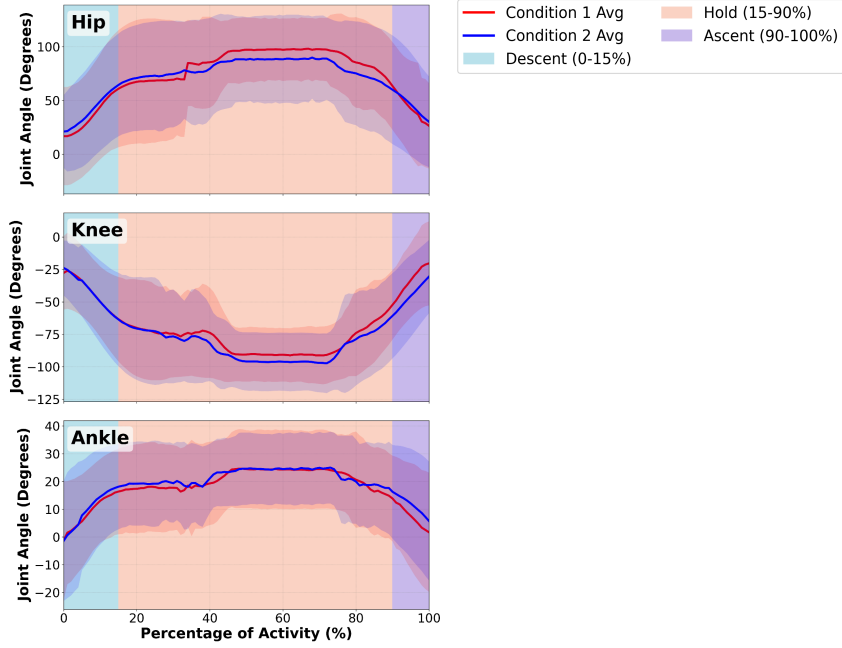
This study evaluated whether orthopedic inserts alter lower-extremity biomechanics during squatting. No significant differences in ankle, knee, or hip joint angles or torques were observed between conditions, suggesting that inserts did not substantially affect squat mechanics in healthy adults. Minor torque differences during the hold phase may reflect subtle compensatory changes in joint loading strategy.

Strengths of this study included the use of motion capture, force plate data, OpenSim modeling, and a within-subject design. However, the small sample size, model-based motion reconstruction, and lack of direct muscle activation measurements limited the robustness of the findings.

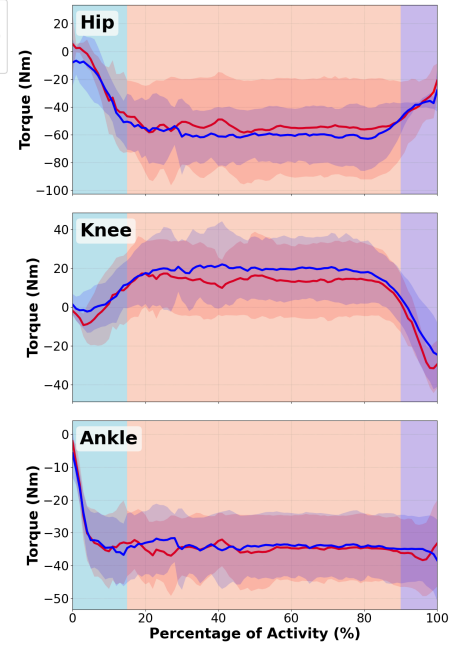
Future studies should include larger sample sizes, electromyography (EMG), and additional balance or center-of-pressure analyses to better evaluate how orthopedic inserts influence lower-extremity biomechanics.

Appendix A.

Bilateral Average Joint Angles with ± 1 SD Cloud



Bilateral Average Joint Torques with ± 1 SD Cloud



Works Cited

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6. Elvitigala, D. S., Matthies, D. J. C., David, L., Weerasinghe, C., & Nanayakkara, S. (2019). GymSoles: Improving squats and dead-lifts by visualizing the user's center of pressure. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–12). ACM. <https://doi.org/10.1145/3290605.3300404>